

Реляционная модель

Атрибут

- У каждого атрибута должно быть уникальное имя в пределах своей таблицы
- В каждом кортеже содержится значение атрибута, которое принадлежит домену атрибута.

Реляционная модель

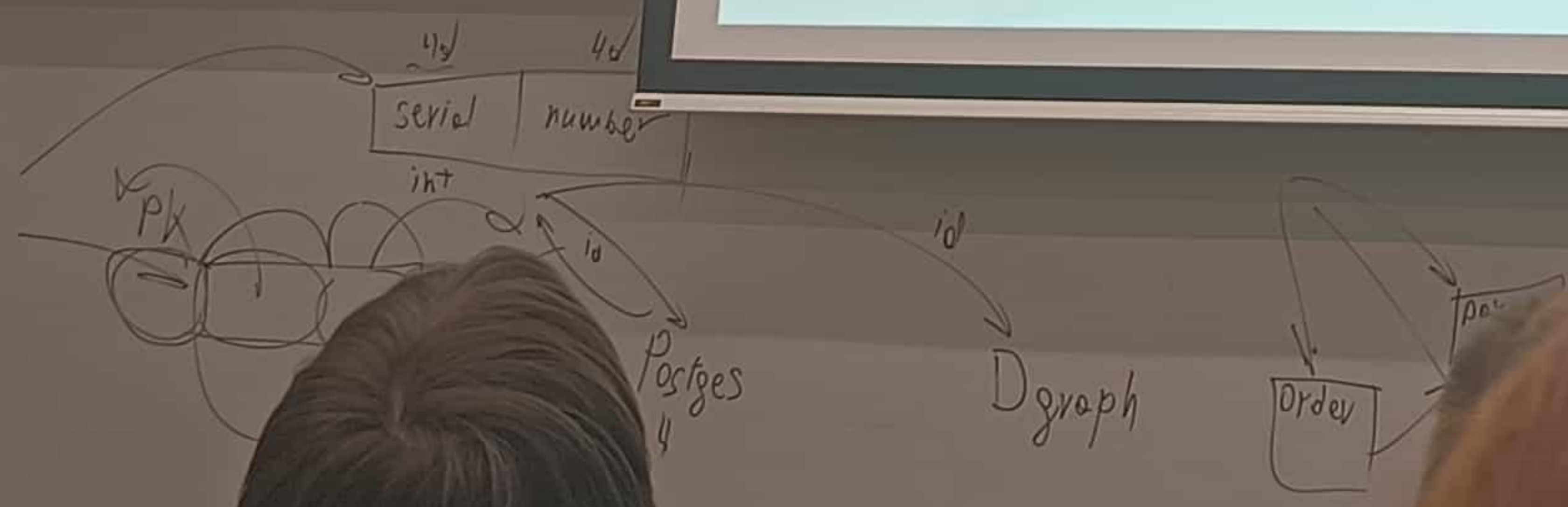
Виды атрибутов

- Простой
- Составной
- Однозначный
- Многозначный
- Производный

Реляционная модель

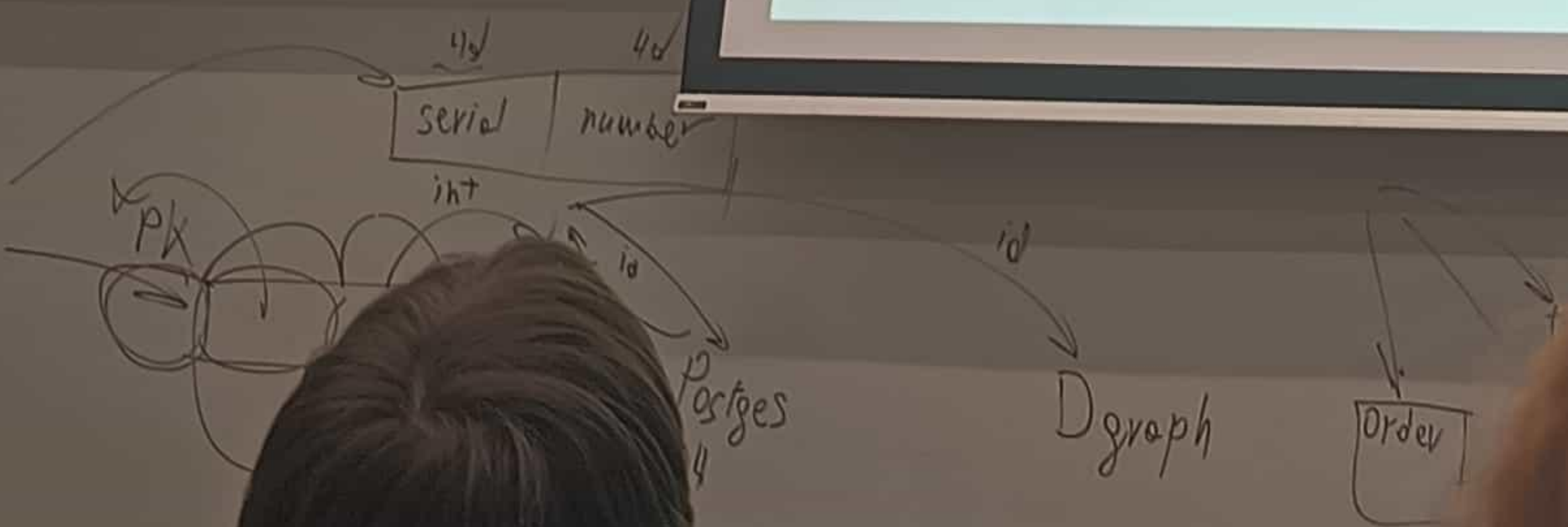
Типы данных

- Целочисленные типы (smallint, int, bigint)
- С плавающей точкой (real, double)
- Строки (varchar, char, text)
- Дата и время (date, time, timestamp, interval)
- Логический (bool)
- Геометрические типы (point, path, line)
- json



Реляционная модель Null

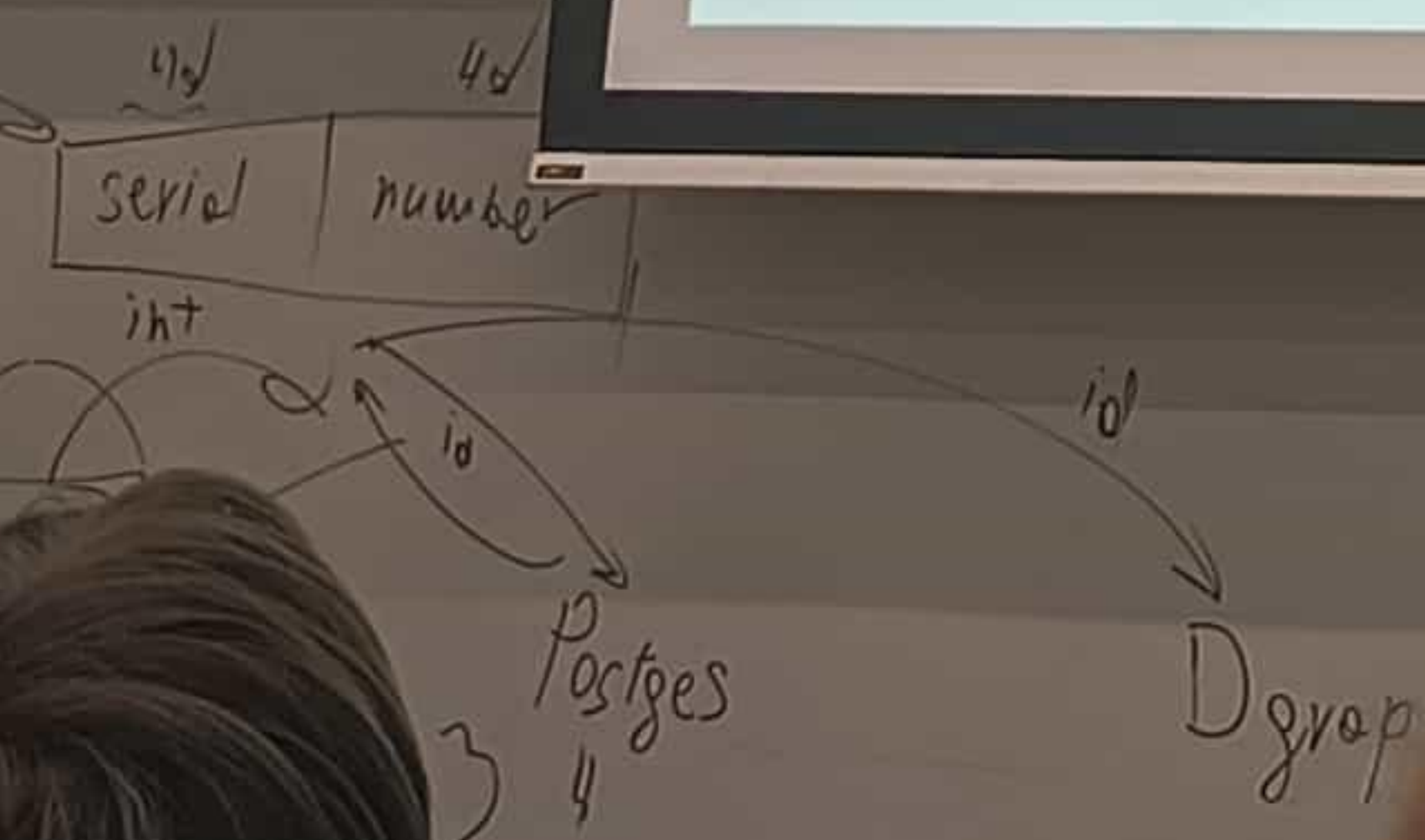
- `SELECT 1 = NULL` вернет NULL
- `1 <> NULL` вернет NULL
- `SELECT 1 IS NULL` вернет 0
- `NULL IS NULL` вернет 1



Реляционная модель

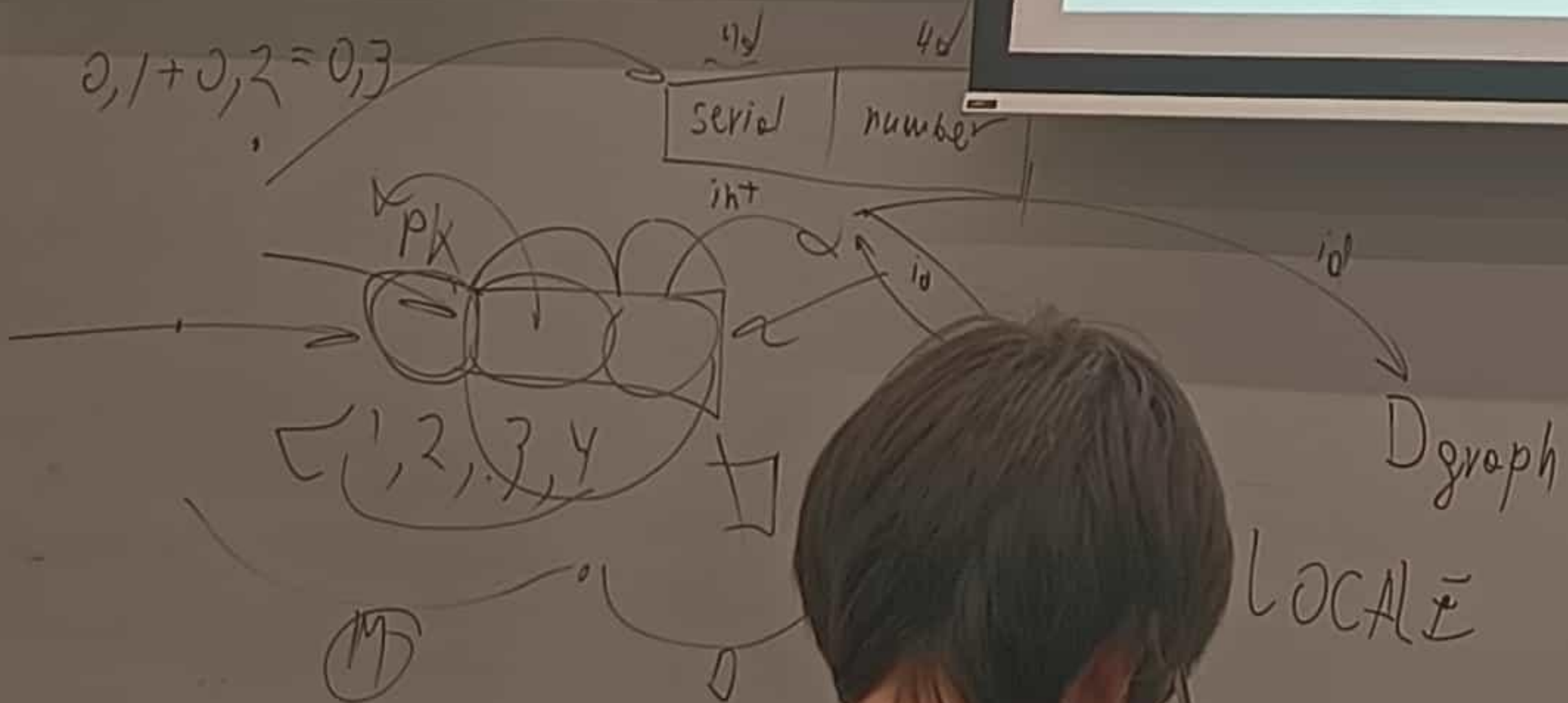
Числовые типы

- Фиксированные (SMALLINT, INT, BIGINT)
- DECIMAL(M, D)
- Плавающая точка (REAL, DOUBLE)
- Денежный
- Авотинкрементные (SMALLSERIAL, SERIAL, BIGSERIAL)



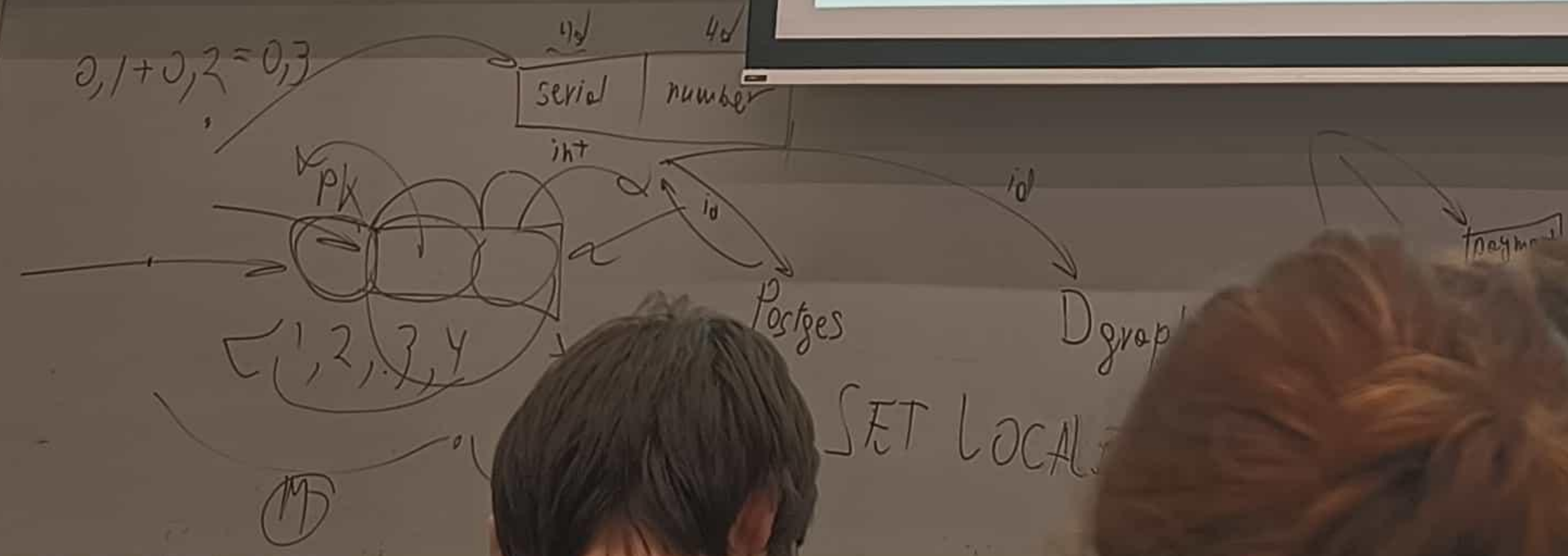
DML

- **SELECT** – выборка данных
- **INSERT** – вставка новых данных
- **UPDATE** – обновление данных
- **DELETE** – удаление данных
- **MERGE** – слияние данных



DDL

- CREATE TABLE
- ALTER TABLE
- DROP
- TRUNCATE



SELECT

Кусочек документации PostgreSQL 16

```
[ WITH [ RECURSIVE ] with_query [, ...] ]
SELECT [ ALL | DISTINCT [ ON ( expression [, ...] ) ] ]
    [ * | expression [ [ AS ] output_name ] [, ...] ]
    [ FROM from_item [, ...] ]
    [ WHERE condition ]
    [ GROUP BY [ ALL | DISTINCT ] grouping_element [, ...] ]
    [ HAVING condition ]
    [ WINDOW window_name AS ( window_definition ) [, ...] ]
    [ { UNION | INTERSECT | EXCEPT } [ ALL | DISTINCT ] select ]
    [ ORDER BY expression [ ASC | DESC | USING operator ] [ NULLS
{ FIRST | LAST } ] [, ...] ]
    [ LIMIT { count | ALL } ]
    [ OFFSET start [ ROW | ROWS ] ]
    [ FETCH { FIRST | NEXT } [ count ] { ROW | ROWS } { ONLY | WITH
TIES } ]
    [ FOR { UPDATE | NO KEY UPDATE | SHARE | KEY SHARE }
[ OF table_name [, ...] ] [ NOWAIT | SKIP LOCKED ] [...] ]
```


FROM

Выбор таблицы

```
SELECT *  
FROM SCHEMA.TABLE
```

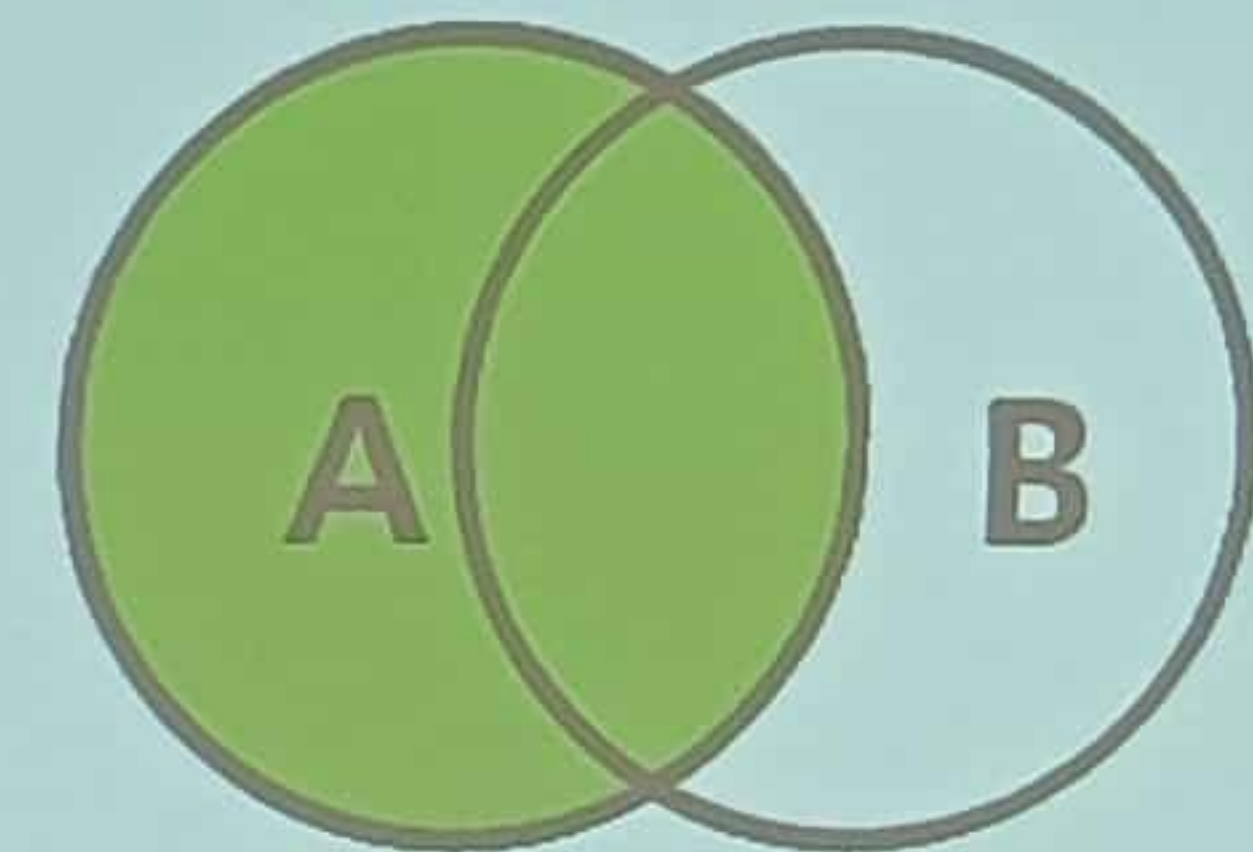
```
SELECT {a1, a2}  
FROM SCHEMA.TABLE
```

```
SELECT * FROM (SELECT id, name FROM employees) AS emp_sub;
```

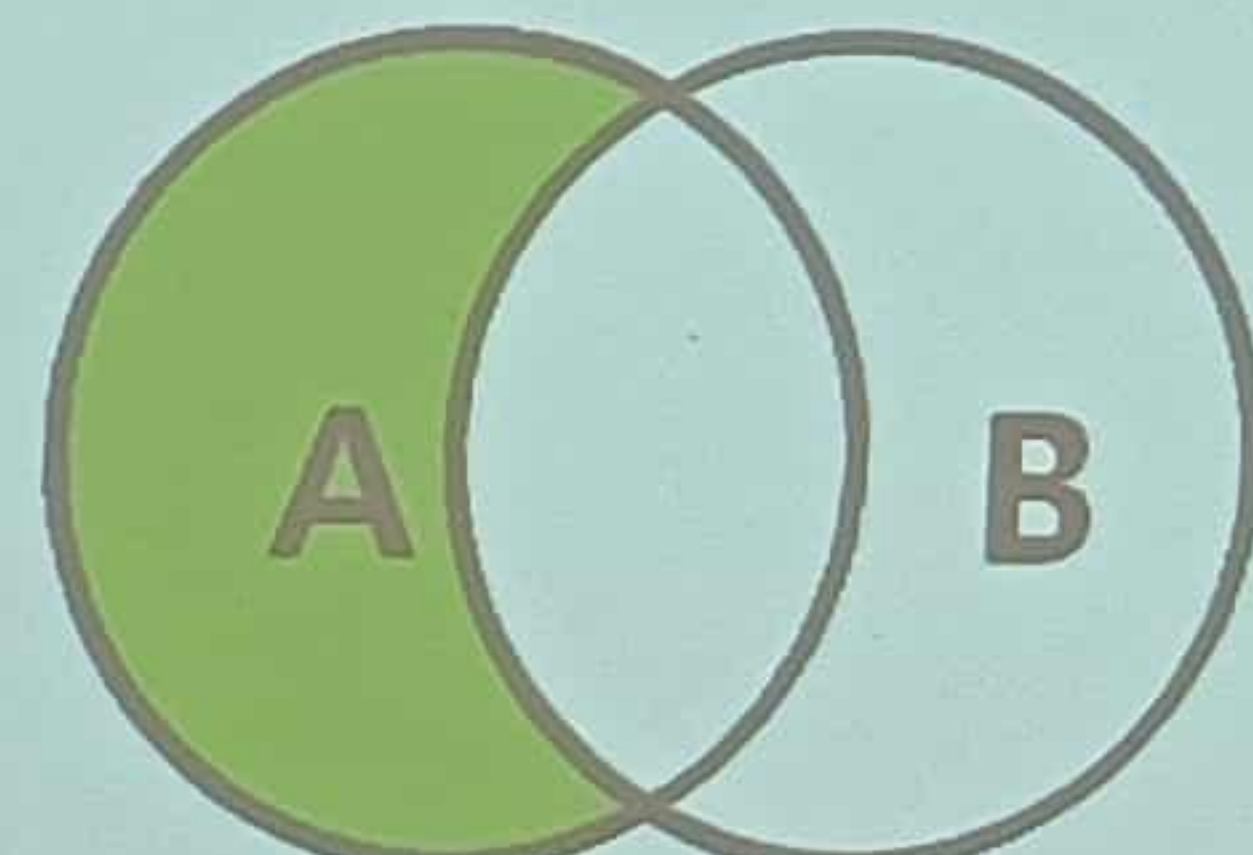


JOIN

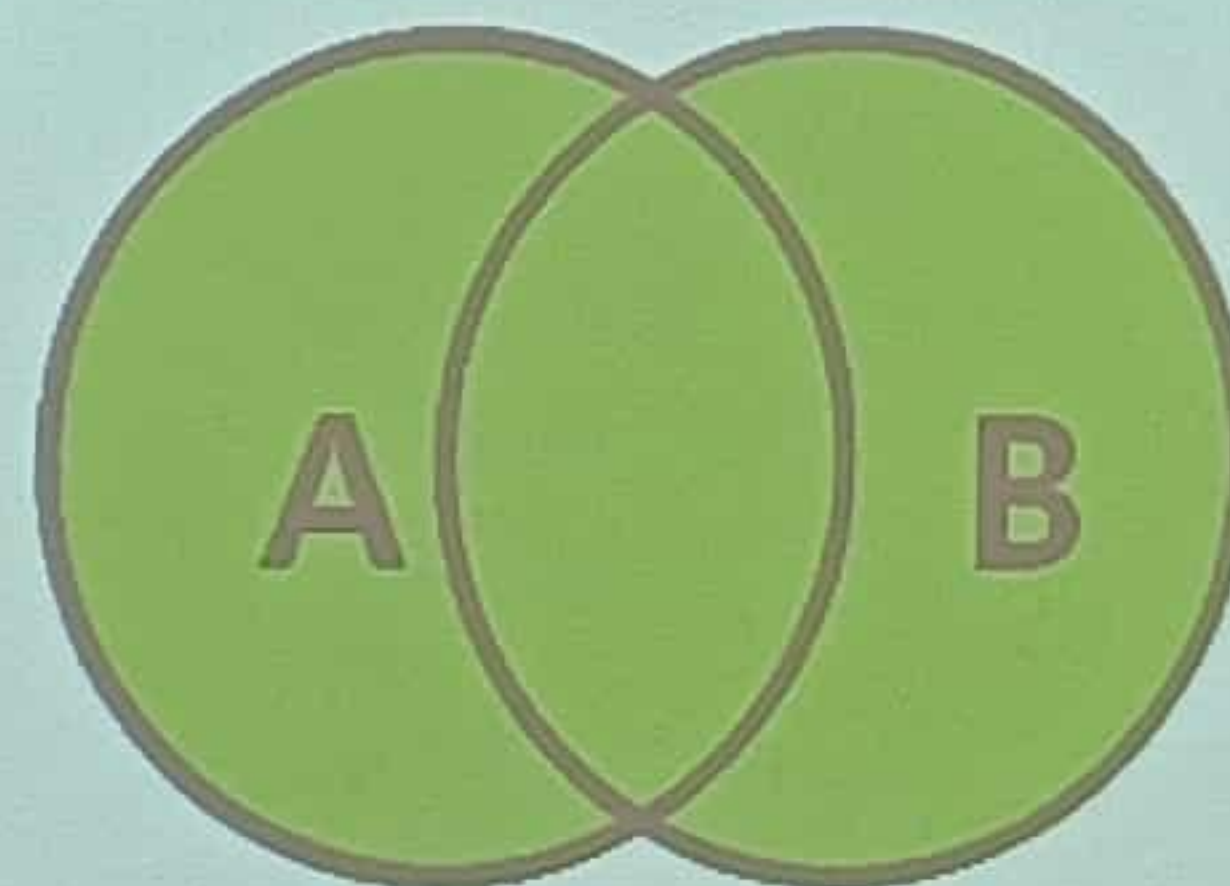
Объединение



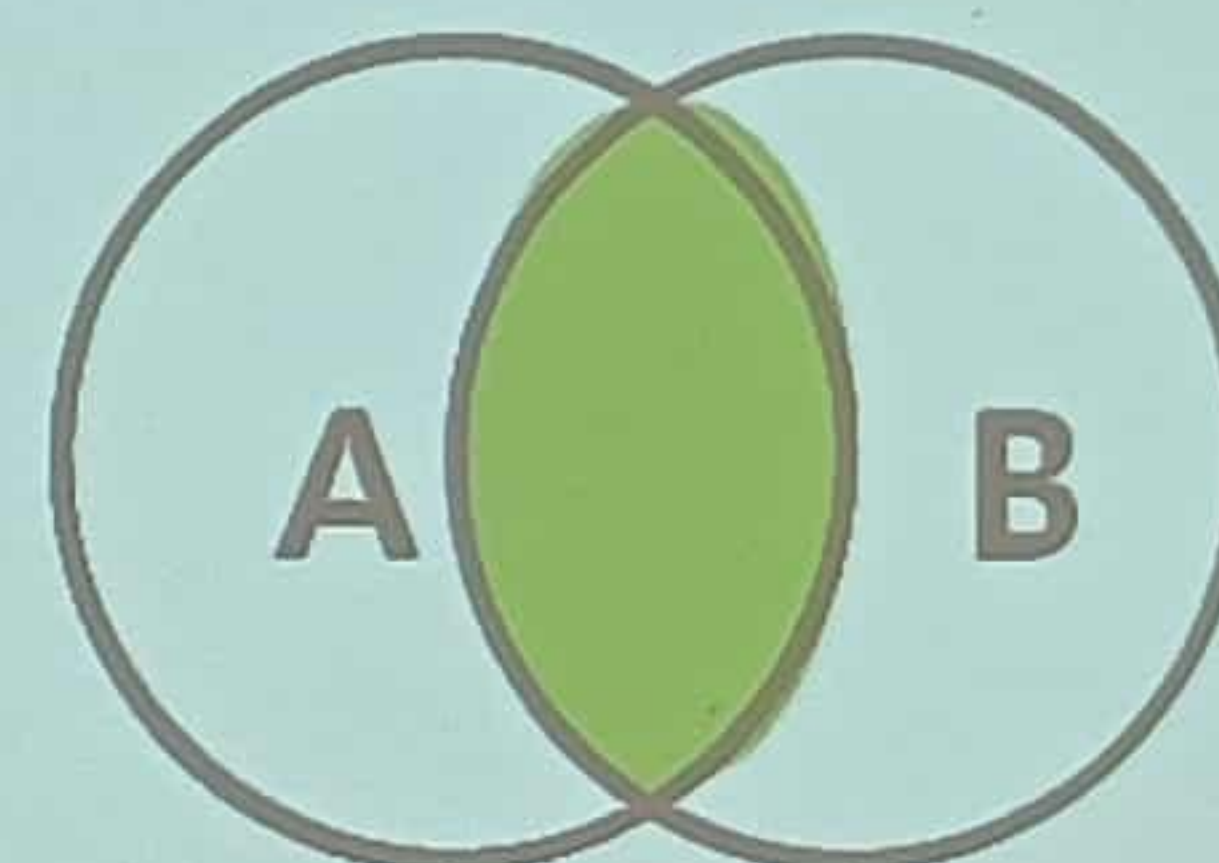
```
SELECT * FROM TableA A  
LEFT JOIN TableB B  
ON A.key = B.key;
```



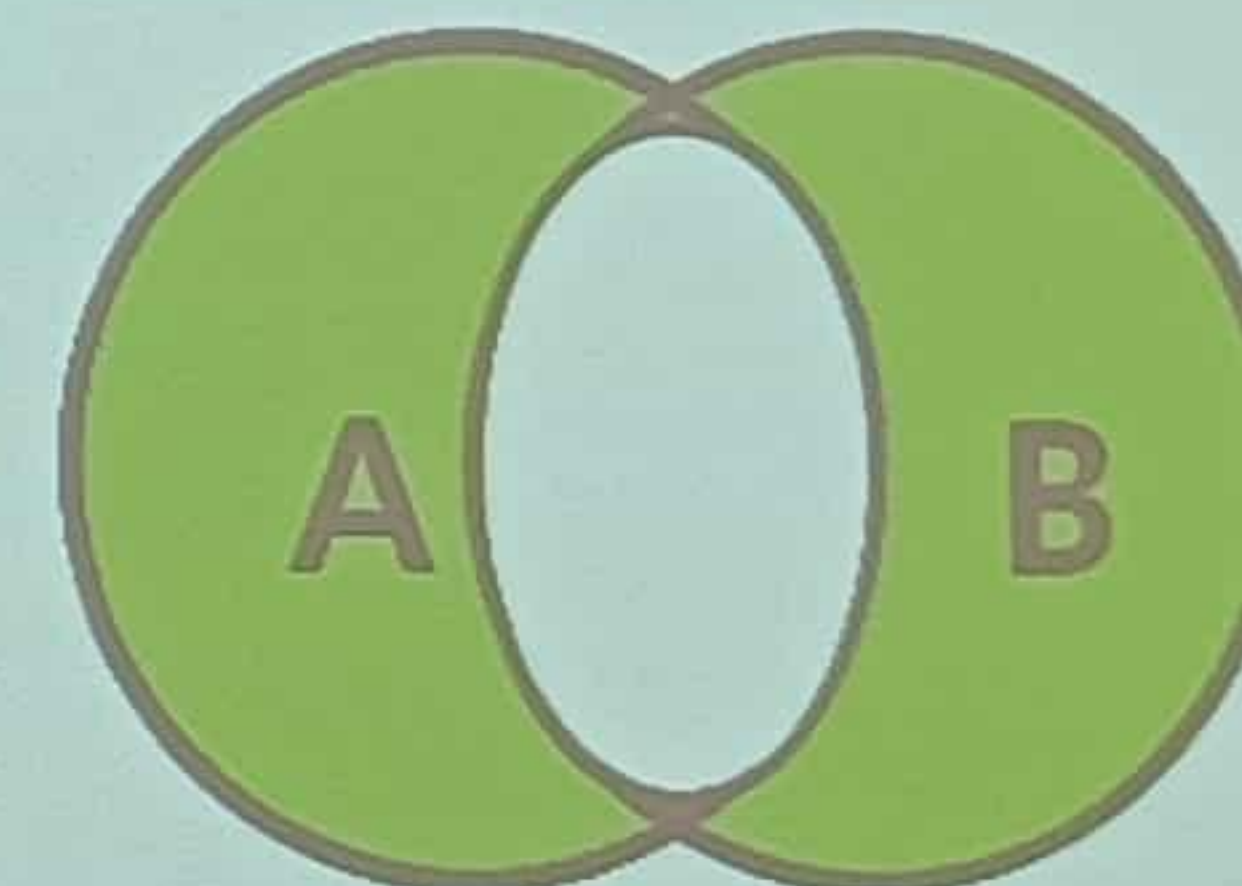
```
SELECT * FROM TableA A  
LEFT JOIN TableB B  
ON A.key = B.key  
WHERE B.Key IS NULL;
```



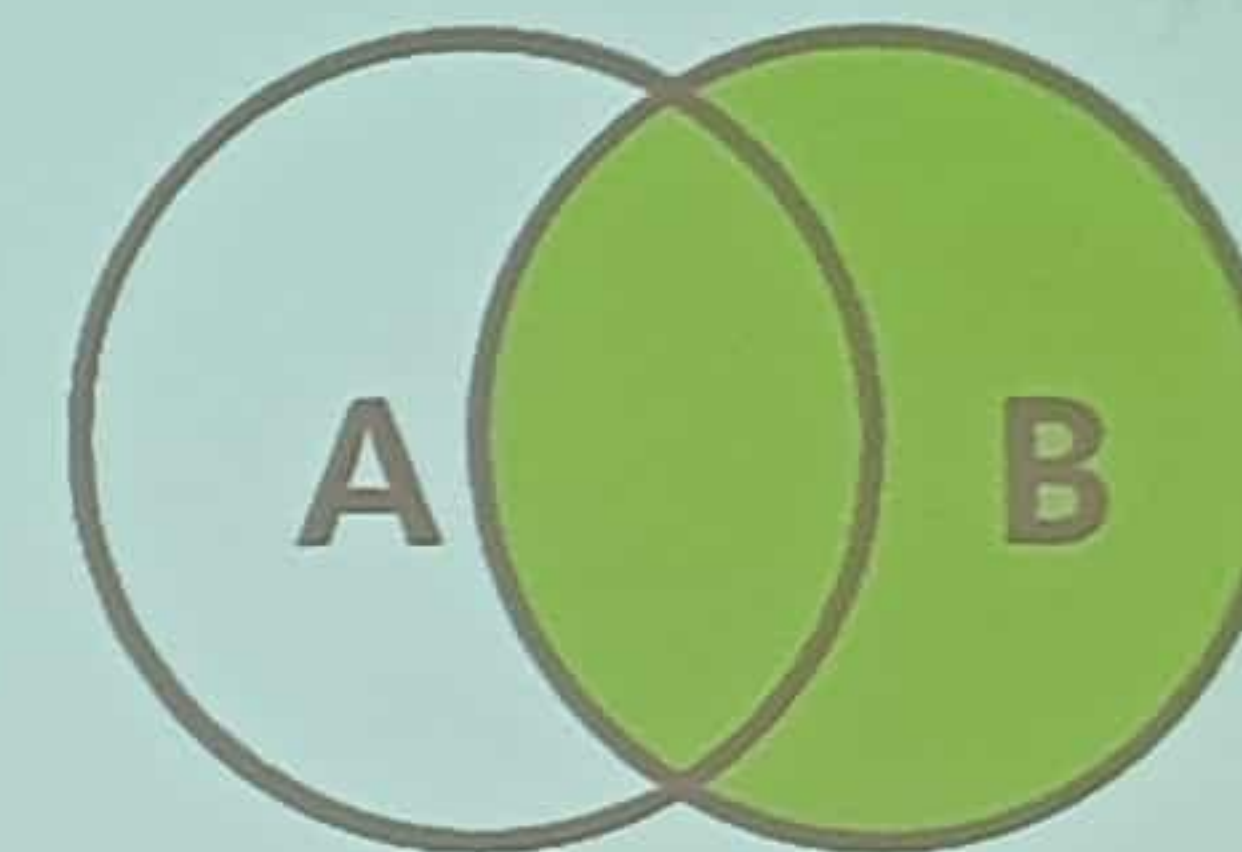
```
SELECT * FROM TableA A LEFT JOIN TableB B  
UNION  
SELECT * FROM TableA A RIGHT JOIN TableB B
```



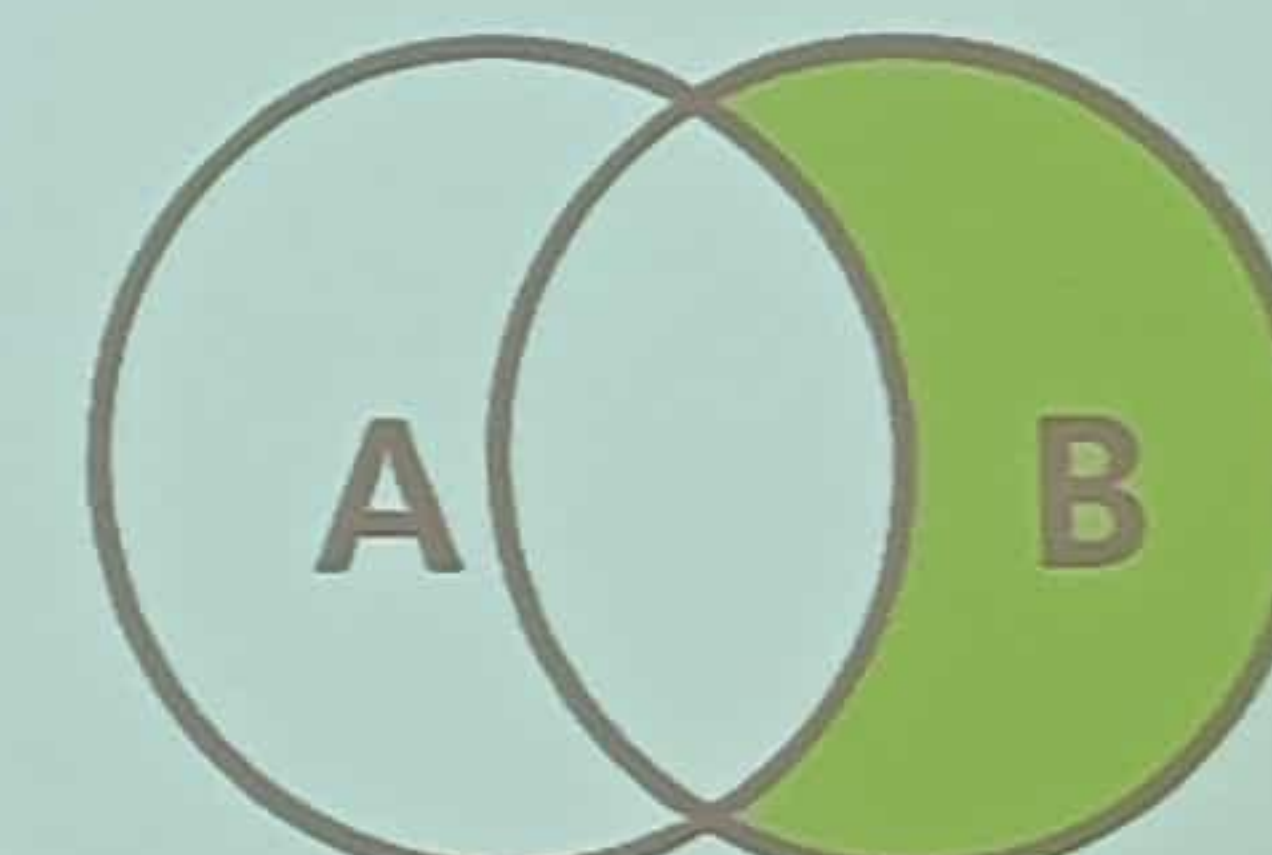
```
SELECT * FROM TableA A INNER JOIN TableB B ON A.key = B.key;
```



```
SELECT * FROM TableA A LEFT JOIN TableB B  
UNION  
SELECT * FROM TableA A RIGHT JOIN TableB B  
WHERE A.key IS NULL OR B.key IS NULL
```



```
SELECT * FROM TableA A  
RIGHT JOIN TableB B  
ON A.key = B.key;
```



```
SELECT * FROM TableA A  
RIGHT JOIN TableB B  
ON A.key = B.key  
WHERE A.Key IS NULL;
```